Carbon-Aware Scheduling for Phase-Disaggregated LLM Inference in Cloud and Serverless Platforms: Review and Simulation

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***Abstract*—Interactive large language model (LLM) serving separates into compute-intensive prefill and memory-bound decode stages across geo-distributed cloud clusters and bursty serverless GPU pools. A systematic PRISMA-2020 synthesis of 47 studies shows that zero prior phase-disaggregated LLM engines (0/6) incorporate electricity grid carbon intensity or amortized embodied hardware emissions. Conversely, zero multi-region cloud carbon schedulers (0/4) model key-value (KV) cache wide-area transfer economics, and 78.7% of reviewed works omit manufacturing carbon. To address these gaps, I formalize a dual-track control architecture combining ISO/IEC 21031:2024 Software Carbon Intensity (SCI) accounting, hybrid carbon-and-queue-aware phase routing across colocated, intra-region, cross-region KV-gated, and remote execution paths, and hazard-driven dynamic serverless keep-alive control. Simulations in SimPy 4.1.2 over 10 seeds using Azure conversational and coding LLM traces alongside UK NESO and Japan TSO grid telemetry show that hybrid routing lowers operational SCI by up to 60.8% and total SCI by up to 54.9% on conversational traffic at 0.81%–0.82% SLO violations, while cutting bursty-code tail SLO violations by up to 86.0% below unconditional geographic disaggregation. On wind-clean grids, embodied emissions reach 93.0%–97.6% of total SCI, causing clean-region over-provisioning to raise total lifecycle carbon by 27.7%–32.6%.**

***Index Terms*—Carbon-aware scheduling, embodied carbon, large language model inference, phase disaggregation, serverless GPUs, Software Carbon Intensity.**

I. Introduction

Large language model (LLM) inference has shifted datacenter energy demand from offline batch training toward continuous, latency-critical serving. Recent survey estimates indicate that U.S. datacenter electricity consumption could climb from 176 TW h in 2023 toward 580 TW h by 2028 [1]. Because interactive LLM serving runs across globally distributed cloud clusters, its carbon footprint depends on compute power draw, temporal and regional electricity grid carbon intensity (gCO_2eq/kWh), and hardware manufacturing emissions amortized over server lifetimes [2], [3].

Two distinct research communities have addressed pieces of this problem over the past five years, yet their methods have developed largely in isolation. First, systems researchers have shown that autoregressive LLM generation splits naturally into two phases with contrasting bottlenecks [4]–[6]: a compute-bound *prefill* phase that processes prompt tokens in parallel to satisfy time-to-first-token (TTFT) bounds, and a memory-bandwidth-bound *decode* phase that streams key-value (KV) caches autoregressively under tight time-per-output-token (TPOT) constraints. Disaggregating prefill and decode across separate GPU pools eliminates phase interference and allows aggressive power capping or lower-tier GPUs on decode pools [4], [7], improving inference energy efficiency at the node or cluster level [8], [9]. Second, sustainable cloud researchers have designed spatiotemporal workload schedulers and carbon-aware serverless keep-alive controllers that follow clean grid hours or low-carbon regions [10]–[14].

A systematic audit of 47 retained works (38 non-survey studies and 9 surveys) reveals sharp architectural gaps between these communities. Zero of the six phase-disaggregated LLM serving designs (0/6) [4], [5], [8], [15]–[17] incorporate dynamic grid carbon intensity signals or amortized hardware embodied emissions. Conversely, zero of the four cross-region carbon-aware cloud schedulers (0/4) [12], [13], [18], [19] account for LLM prefill/decode disaggregation or the bandwidth and energy cost of migrating large KV caches across wide-area networks (WANs).

Furthermore, 37 of the 47 reviewed publications (78.7%) omit hardware embodied carbon entirely, even though lifecycle studies show that manufacturing emissions reach 24%–35% of LLM operational totals on average grids [20] and dominate net emissions once operational grid carbon approaches zero [2], [21]. Likewise, serverless cold-start carbon studies target lightweight CPU or small containerized functions (< 15 s startup) [14], [22], [23]; as a result, multi-gigabyte LLM weights (10.3 s to 84.0 s reload times for 70B parameters [24]) remain largely unexamined under dynamic carbon-aware keep-alive policies.

This work bridges both domains through a systematic PRISMA-2020 review and a trace-driven discrete-event simulation study structured around four research questions:

- **RQ1 (Taxonomy and Architectural Gaps):** How do existing carbon-aware cloud schedulers, serverless keep-alive controllers, and phase-disaggregated LLM serving systems align across grid signal granularity, phase awareness, and embodied amortization?

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- **RQ2 (Disaggregated Carbon–SLO Frontier):** Under real regional grid carbon-intensity traces and production LLM request traces, can hybrid carbon-and-queue-aware phase routing reduce operational and lifecycle Software Carbon Intensity (SCI) relative to latency-only colocation and phase-agnostic carbon routing without violating TTFT and TPOT SLOs?
- **RQ3 (Serverless Keep-Alive Economics):** Across clean and carbon-intensive grids, when does hazard-driven dynamic keep-alive cut operational carbon versus static timeouts without triggering severe cold-start SLO violations, and how strongly does checkpoint loader latency shift that break-even point?
- **RQ4 (Embodied Scale-Out Paradox):** When does adding spare GPU nodes in low-carbon regions invert net lifecycle carbon (SCI_{tot}), and how do decoupled host versus GPU lifetimes (9 yr/3 yr vs. 4 yr) change cluster provisioning choices?

This paper makes three contributions:

1) **Dual-Axis Systematic Synthesis:** Following PRISMA-2020 [25], I screen 75 candidate records into a structured 47-work corpus (42 full-text, 5 abstract-only) covering 35 empirical works, 9 surveys, 2 standards, and 1 trace release. The taxonomy identifies exact empty design intersections (0/6 and 0/4) alongside seven open challenges (`OC1–OC7`).
2) **Carbon-Extended Disaggregated & Serverless Model:** I formalize an ISO/IEC 21031:2024 [3] lifecycle model inside a discrete-event SimPy 4.1.2 simulator parameterized with published GPU power/throughput curves [4], [7], GreenSKU/LLMCarbon embodied profiles [20], [21], WAN KV-cache crossover gates, and hazard-based serverless idle control [14], [24].
3) **Multi-Seed Empirical Evaluation (`E1`, `E2`, `SW1–SW4`, `H4`):** Across 10 random seeds per condition (Student-t 95% CIs and paired Wilcoxon tests) driven by CC-BY-4.0 Azure LLM traces [26], UK NESO, and Japan TSO telemetry, hybrid routing (`P4`) cuts operational SCI by up to 60.8% (-54.6%–54.9% total SCI, $p = 0.002$) at 0.81%–0.82% SLO violations on conversational traffic, reduces bursty-code tail SLO violations by 85.2%–86.0% over phase-agnostic carbon routing, finds a 97.6% embodied floor in clean serverless grids (`South Scotland`), and shows that adding four spare nodes in clean regions raises total SCI by 27.7%–32.6%.

The remainder of the paper is organized as follows. Section II reviews LLM serving phases, WAN KV bounds, GPU cold starts, and ISO/IEC 21031:2024 accounting. Section III details the PRISMA-2020 protocol and corpus audit. Section IV answers `RQ1` with a comparative taxonomy. Section V specifies the parameterized simulation scope, equations, traces, and statistical protocol. Section VI evaluates `E1`, `E2`, `SW1–SW4`, and `H4_elastic` and answers `RQ1–RQ4`. Section VII synthesizes open challenges and limitations, and Section VIII concludes with Generative AI and data availability statements.

II. Background and Preliminaries

Building carbon-aware LLM serving schedulers requires joining four domains that developed under separate abstractions: phase-level transformer execution, wide-area carbon routing, GPU serverless cold-start management, and lifecycle carbon accounting under ISO/IEC 21031:2024. Fig. 1 illustrates how these layers interact inside the dual-track control architecture.

A. Autoregressive LLM Phases and Disaggregation

Within multi-tenant cloud platforms [27], autoregressive LLM serving proceeds in two stages for each request with prompt length L_{in} and generation length L_{out}. During *prefill*, the model processes all L_{in} prompt tokens concurrently through matrix multiplications to produce the first token and populate the KV cache, which determines time-to-first-token (TTFT). During *decode*, the engine iterates token by token over L_{out} steps, reloading model weights and the growing KV cache from high-bandwidth memory (HBM) on each step; this saturates memory bandwidth before arithmetic units and governs time-per-output-token (TPOT).

Production traces show strong workload skew across domains: Azure LLM logs report conversational prompts averaging $1{,}154.7$ input and 211.1 output tokens with steady arrivals (coefficient of variation CV $= 1.094$), whereas coding completions average $2{,}047.8$ input and 27.9 output tokens under extreme burstiness (CV $= 13.152$) [26]. Mooncake traces exhibit even heavier prompt skew, averaging $8{,}590$ input versus 182 output tokens [15].

Local memory and batching schedulers address single-cluster bottlenecks first. PagedAttention in vLLM cuts KV waste from 61.8%–79.6% down to 3.7% and saves up to 55.2% KV memory via copy-on-write [28], while Sarathi-Serve uses chunked prefill and stall-free batching to eliminate 28.3× generation stalls [6]. Even with chunked batching, colocated requests couple prefill latency spikes with decode tail violations and restrict hardware specialization.

Phase-disaggregated architectures (Splitwise [4], DistServe [5], Mooncake [15], and DualScale [8]) separate prefill and decode onto distinct GPU pools connected via NVLink or RDMA fabrics. Splitwise shows that decode GPUs tolerate a 50% power-cap drop (700 W down to 350 W on H100) with minimal TPOT impact while achieving 2.35× iso-power-cost throughput [4]; DualScale cuts prefill energy by 39% and decode energy by 48% at P99 latency parity [8]. Empirical measurements confirm that decode still dominates cumulative energy across token ratios [9], and bilinear $(L_{\text{in}}, L_{\text{out}}, L_{\text{in}}L_{\text{out}})$ energy regression achieves $R^2 \geq 0.964$ [29]. However, recent dual-GPU experiments by Li et al. [16] caution that disaggregation can underperform colocated inference at small batch sizes over constrained interconnects until batch-32 KV eviction pressure triggers colocation collapse.

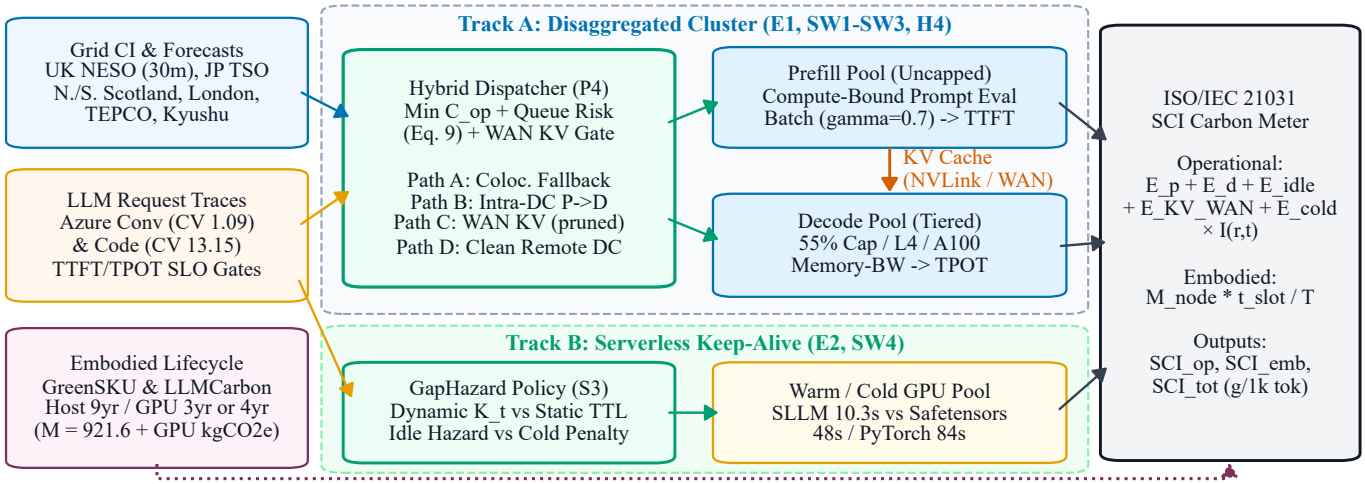


Fig. 1. Dual-track carbon-aware phase-disaggregated LLM cluster (Track A: Paths A–D via Eq. (9)) and serverless GPU keep-alive (Track B: GapHazard policy `S3`) feeding the unified ISO/IEC 21031:2024 SCI telemetry meter.

B. Wide-Area Carbon Routing and KV Transfer Economics

Regional electricity grids vary widely in hourly operational carbon intensity $I(r,t)$ (gCO_2eq/kWh). Across 123 Electricity Maps grids, Sukprasert et al. [12] report a 40× spread (mean 368.39 gCO_2eq/kWh) and note that 87% of 40 sampled datacenter grids exhibit 24-hour periodicity. Yet latency and capacity constraints sharply curtail pure carbon-chasing: unconstrained 96% spatial reductions shrink to 51.5% under 50% regional spare capacity and 31% under a 50 ms WAN round-trip time (RTT) ceiling, while 50% grid-forecast error introduces 10%–12% overhead [12].

Live multi-region Kubernetes scheduling in GreenCourier cuts carbon by 8.7%–17.8% across four European zones at a 10.3%–16.2% latency penalty [18], whereas single-cluster temporal deferrals [10], elastic scaling [11], battery orchestration [30], and microservice DAG routing [13] rely on delay tolerance or small HTTP payloads. In cloud-to-edge continuums, linear-programmed function variant selection lets 30% more requests satisfy QoS targets and cuts execution carbon by 20% under equal budgets [19].

For LLM serving, splitting prefill and decode across WAN regions adds serial KV-cache transmission: a 70B-class FP16 GQA KV state at long context spans hundreds of megabytes per request, adding both WAN transmission latency (L_{kv}/B_{wan}) and network transfer energy (0.001–0.005 kWh/GB [13]). Cross-region disaggregation is therefore beneficial only when the carbon savings between regions exceed network transfer emissions without breaching TTFT SLOs.

C. Serverless GPU Keep-Alive and Checkpoint Loading

Under bursty traffic, such as Azure Functions where 81% of apps see ≤ 1 req/min [31], idle servers draw static power while eviction incurs cold-start energy. In Kepler benchmarks, idle CPU draw reaches 21%–83% of peak power [14], and an idle GPU draws 60–100 W. For 70B-class LLMs (about 130 GB weights), checkpoint cold-start duration spans 84.0 s under standard PyTorch loading, 48.0 s under Safetensors, and 10.3 s even with ServerlessLLM NVMe RAID-0 Direct I/O [24].

Recent dynamic keep-alive controllers (EcoLife [23], CASA [22], LowCarb [32], LACE-RL at -77.08% idle carbon [14], and Festina at -56% energy and -85% cold starts [17]) show that adaptive idle timeouts outperform fixed windows, yet systematic comparison across multi-gigabyte loader tiers and clean versus coal-heavy grids remains open.

D. ISO/IEC 21031:2024 SCI and Embodied Carbon

To prevent shifting emissions between operational and hardware budgets, the Green Software Foundation Software Carbon Intensity (SCI) specification, standardized as ISO/IEC 21031:2024 [3], mandates an additive lifecycle intensity per functional unit R (here, 1,000 processed tokens) while explicitly excluding offset purchases:

$$\mathrm{SCI} = \frac{(E \cdot I) + M}{R}, \tag{1}$$

where E is total consumed electricity (kWh, inclusive of datacenter PUE), I is regional grid carbon intensity (gCO_2eq/kWh), and $M = M_{\text{server}} \cdot (t_{\text{slot}}/T_{\text{life}})$ is amortized manufacturing carbon.

Industry GHG audits [2], architectural models [33], [34], and LLM carbon tools [20], [21] show that embodied emissions (800 $kgCO_2eq$ host baseline [21]; 921.6 $kgCO_2eq$ in the GreenSKU Genoa assembly, plus die/HBM [20]) represent 24%–35% of LLM totals on average grids and surpass 80% as grids decarbonize. Consequently, metrics using raw GPU utilization [35] or ignoring power capping [7], [36], [37] and aging amortization [38], [39] risk optimizing local Watts while raising system-wide lifecycle carbon. These tradeoffs motivate unified simulation testbeds in the tradition of CloudSim [40].

III. PRISMA-2020 Review Methodology

To ground RQ1 and parameterize the simulation environment from primary sources, I conducted a systematic literature review following PRISMA-2020 reporting principles across academic indices (Crossref, Semantic Scholar, OpenAlex, arXiv, ACM Digital Library, IEEE Xplore, USENIX, and official standards repositories).

A. Search Queries, Inclusion Criteria, and Screening

Candidate studies published between 2011 and 2026 were retrieved through three structured search tracks targeting the intersection of sustainable cloud computing, LLM inference disaggregation, and serverless GPU control:

- **Query A (Carbon-Aware Cloud & Serverless Scheduling):** Paired carbon-intensity, grid emissions, SCI, and embodied terms with spatial/temporal shifting, serverless keep-alive, and cold-start mitigation (16 primary records).
- **Query B (Phase-Disaggregated & Energy-Aware LLM Serving):** Combined LLM inference, KV caching, prefill/decode disaggregation, and GPU serving with energy, DVFS, power capping, and carbon accounting (30 primary records).
- **Query C (Surveys, Traces, Embodied Models & Snowballing):** Aggregated recent PRISMA surveys, lifecycle standards, hardware embodied datasets, and backward/forward citation chasing (29 records).

As traced in Fig. 2, removing 1 exact duplicate reduced the 75 identified records to 74 unique candidates for title and abstract screening. In total, 11 records were excluded during screening for lacking cloud computing, datacenter energy, or LLM execution scope. Full-text and structured metadata eligibility assessment covered 63 records; 16 additional items were excluded because they addressed purely offline model training without serving SLOs or contained neither carbon nor LLM systems evidence.

B. Corpus Provenance, Full-Text Auditing, and Preprint Disclosure

The final synthesis corpus comprises 47 works: 35 empirical or modeling systems, 9 surveys or taxonomies, 2 standards/specifications (ISO/IEC 21031:2024 [3] and NIST SP 800-145 [27]), and 1 production trace release [26]. By publication year, 42 of 47 items (89.4%) appeared between 2022 and 2026, and 23 (48.9%) appeared in 2025–2026.

To enforce strict claim verification, every retained work is tagged by verified retrieval depth and peer-review status:

1) **Full-Text Audited (42/47, 89.4%):** Complete PDFs or official texts underwent page, table, and equation verification: 34 archival items (31 papers, 2 standards, and 1 trace release), plus 8 preprints: six arXiv preprints (Hewage et al. [41], Yang et al. [1], Pan et al. [42], EcoServe [21], DualScale [8], and Li et al. [16]) and two accepted conference preprints (LACE-RL [14] for CCGrid 2026 and Festina [17] for ACM SIGOPS ATC 2026).
2) **Abstract-Only Verified (5/47, 10.6%):** Paywalled works limited to publisher abstracts and metadata (LowCarb [32], Calavaro et al. [19], ACT [33], Sinha et al. [35], and CloudSim [40]). Claims citing these five records are bounded strictly to verified abstract statements, with zero unverified table constants drawn into the simulator.

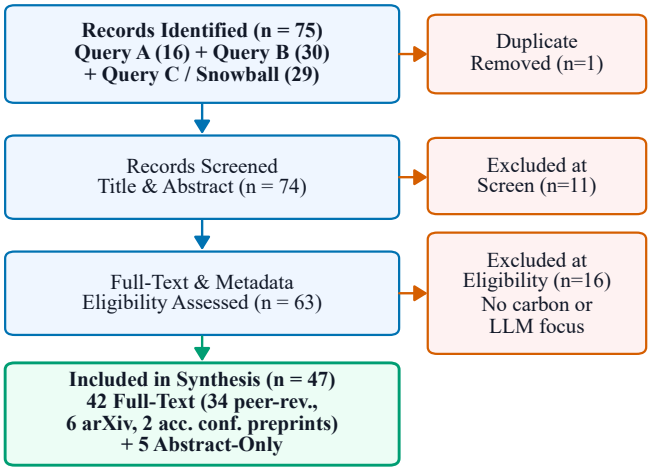


Fig. 2. PRISMA-2020 systematic search, screening, and eligibility pipeline yielding the 47-work corpus (42 full-text, 5 abstract-only).

IV. Taxonomy and Literature Synthesis

This section answers RQ1 by organizing the 47 retained works into six functional classes (L1.A–L1.F) and mapping empirical intersections across phase awareness and carbon/energy signals.

A. Dual-Axis Taxonomy (L1.A–L1.F)

1) L1.A Prior Surveys and Taxonomies ($n = 9$): Recent surveys cover cloud carbon scheduling or LLM efficiency separately without documenting the combined design space (Table I). Hewage et al. [41] synthesize operational and embodied carbon control for latency-sensitive clouds without mentioning LLMs, prefill/decode, or KV caching. Yang et al. [1] review container orchestration across serverless and GPU workloads while omitting disaggregation and embodied carbon. Alam et al. [25] report that 70% (44/63) of surveyed carbon-aware cloud studies rely solely on simulation and only 6.3% (4/63) track live grid intensity. Among serverless reviews, Agarwal et al. [43] find sustainability metrics in only 4 of 40 dynamic resource schemes, and Besozzi et al. [44] report 29.5% GPU support across 122 studies alongside only two carbon mentions.

Conversely, LLM serving surveys by Miao et al. [45], Jiang et al. [46], and Pan et al. [42] examine KV transfer and disaggregation without operational or embodied carbon accounting,

whereas Bai et al. [47] cover operational PUE·kWh·CI metrics without phase disaggregation or embodied manufacturing budgets.

2) L1.B Spatial and Temporal Cloud Carbon Schedulers ($n = 8$): Temporal delay schedulers such as Wait Awhile (-4.1% to -11.2% CO_2 within ± 8 h windows) [10], CarbonScaler (-19% mean AWS savings, -36% Ontario, -51% NL) [11], EcoVisor (-50.05% BLAST via batteries) [30], and Clover ($> 75\%$ CO_2 cut on MIG-partitioned vision/BERT jobs) [39] target batch elasticity or multi-second slacks incompatible with interactive token streaming.

Multi-region frameworks including GreenCourier (-8.7% to -17.8% carbon at 10.3%–16.2% latency increase) [18], Caribou (-22.9% to -66.6% for FaaS DAGs) [13], Calavaro et al. [19] (LP edge-cloud offloading yielding +30% QoS and -20% carbon under equal budgets, verified from abstract only), and Sukprasert et al. [12] show regional carbon arbitrage (40x grid spread) alongside RTT and capacity bottlenecks (dropping from a 96% to a 31% ceiling under 50 ms WAN RTT). None of the four cross-region schedulers (0/4) models LLM P/D disaggregation or WAN KV serialization.

3) L1.C Carbon-Aware Serverless Keep-Alive and Loaders ($n = 7$): Trace characterization on Azure Functions (81% of apps invoked $\leq 1/\text{min}$) [31] motivated dynamic keep-alive controllers. EcoLife cuts idle and operational emissions on multi-generation CPUs within 5.5% of oracle [23]; CASA jointly balances water and grid CO_2 around 15 s Knative cold starts (1.9x–2.8x reduction) [22]; LowCarb applies RL keep-alive control within 15% of optimal (abstract only) [32]; and LACE-RL uses Kepler idle-power ratios (0.21–0.83) to cut cold starts by 51.7% and idle carbon by 77.08% [14].

On GPUs, ServerlessLLM shrinks 70B checkpoint loads to 10.3 s via NVMe Direct I/O (48.0 s Safetensors, 84.0 s PyTorch) [24], while Festina integrates 3D lookup tables, GPU MPS, DVFS, and early-warning eviction (-56% energy, -85% cold starts) alongside a P/D-disaggregated variant (-7% energy) [17]. Yet neither ServerlessLLM nor Festina evaluates large-checkpoint keep-alive dynamics under grid-intensity extremes.

4) L1.D Phase-Disaggregated and Engine-Level LLM Serving ($n = 12$): At the inference engine level, vLLM (cuts KV waste from 61.8%–79.6% to 3.7%) [28], Sarathi-Serve (chunked prefill) [6], Wilkins et al. (bilinear $(L_{\text{in}}, L_{\text{out}})$ energy modeling, $R^2 \geq 0.964$) [29], and ThrottLL'eM (NVML frequency capping up to -43.8% energy) [36] operate without phase disaggregation. Colocated phase-aware governors (DynamoLLM at -38% operational carbon [7], TAPAS at -23%–24% row power and +40% oversubscription [37], and Fernandez et al. at -37.6% to -73.0% energy under vLLM and +55.23% penalty at TP4 [9]) tune local frequencies or cluster capacity on a single grid.

Fully disaggregated systems (Splitwise [4], DistServe [5], Mooncake [15], DualScale at -39% prefill and -48% decode energy [8], and Li et al. [16] on small-batch overhead) optimize intra-datacenter throughput or local Watts/Joules (W/J) while omitting grid CI and embodied emissions (0/6 when counting Festina's P/D variant).

5) L1.E Embodied Hardware and Lifecycle Models ($n = 8$): Lifecycle studies anchor Scope 3 amortization. Gupta et al. [2], [33] show manufacturing dominates once clean grids exceed 80% renewables and formalize Reduce/Reuse/Recycle modeling (ACT verified abstract only). CarbonExplorer shows that 100% annual RECs deliver only 46%–51% hourly carbon-free energy [34], while ISO/IEC 21031:2024 [3] prohibits offset netting in SCI.

For AI accelerators, LLMCarbon models H100 die and DRAM/SSD embodied emissions (24%–35% of serving total) [20], Sinha et al. challenge linear utilization-carbon assumptions across co-located workloads and token ratios (abstract only) [35], Hewage et al. apply NBTI aging awareness (-37.67% P99 annualized CPU embodied carbon) [38], and EcoServe combines 4R principles (9 yr host / 3 yr GPU lifetimes) for up to 47% total carbon savings without phase disaggregation or cross-region routing [21].

6) L1.F Cloud Standards, Simulation Foundations, and Traces ($n = 3$): NIST SP 800-145 defines multi-tenant resource pooling [27], CloudSim establishes extensible federated cloud simulation [40] (abstract only), and the Azure LLM Inference Trace 2024 supplies `conv` and `code` arrival/token distributions [26].

B. Gap Quantification and Synthesis Matrix

Cross-tabulating the 38 non-survey works in Fig. 3 identifies exact zero-coverage intersections. All five pure disaggregated systems [4], [5], [8], [15], [16] plus Festina's P/D variant [17] lie strictly in the Perf/Telem. (2) and Local W/J (4) columns, so 0 works intersect phase disaggregation with either Grid Op. CI or Op. + Embodied. Similarly, only one colocated phase-aware system (DynamoLLM [7]) responds to grid CI, and none (0/4) includes embodied hardware carbon.

Across the complete 47-work corpus (Fig. 4), 37/47 (78.7%) omit embodied accounting, while empirical evaluations split across physical clusters (14/35, 40.0%), hybrid testbed-plus-simulators (10/35, 28.6%), trace simulators/models (9/35, 25.7%), and abstract-only records (2/35, 5.7%). Table II summarizes how the proposed design integrates these previously disconnected mechanisms.

V. Simulation Architecture and Experimental Design

A. Parameterized Simulation Scope Statement

All empirical results in this study derive from a parameterized discrete-event simulation built in Python using SimPy 4.1.2; no telemetry was collected from physical multi-continent multi-GPU clusters. Consistent with the precedent noted by Alam et al. [25] (where 70% of cloud carbon-scheduling studies evaluate wide-area policies through simulation), discrete-event modeling supports reproducible, multi-seed ablations

TABLE I
AUDIT OF PRIOR SURVEYS AND TAXONOMIES (L1.A, $n = 9$) ACROSS CARBON, EMBODIED, AND LLM DISAGGREGATION AXES

Work	Year	Venue / Status	Primary Scope	Grid CI	Emb.	LLM	P/D & KV	Verified Gap Summary
Hewage et al. [41]	2026	arXiv Preprint (28 pp.)	Latency-sensitive cloud	✓	✓	×	×	0 LLM/P/D/KV hits; cites > 60% CO_2 cut
Yang et al. [1]	2025	arXiv Preprint (35 pp.)	Container orchestration	✓	×	Min.	×	Covers SCI, serverless, GPU; 0 P/D/KV/embodied
Alam et al. [25]	2026	*Computing* (25 pp.)	Cloud carbon RL/heuristic	6.3%	×	×	×	70% (44/63) sim-only; 0 GPU/LLM/embodied
Agarwal et al. [43]	2026	*ACM CSUR* (36 pp.)	Serverless adaptation	4/40	×	Min.	×	Sustainability secondary; 0 P/D/KV/embodied
Besozzi et al. [44]	2025	*IEEE Access* (46 pp.)	Serverless architectures	2 hits	×	×	×	29.5% GPU support; 0 LLM/P/D/embodied
Miao et al. [45]	2025	*ACM CSUR* (35 pp.)	Generative LLM serving	Min.	×	✓	✓	Covers KV & disagg; 0 embodied/CI/water
Bai et al. [47]	2026	*ACM CSUR* (70 pp.)	LLM serving efficiency	✓	×	✓	×	PUE·kWh·CI metrics; 0 disagg/cold/embodied
Jiang et al. [46]	2026	*ACL Findings* (27 pp.)	KV-cache compression	×	×	✓	✓	Recommends J/tok; 0 grid CI or embodied
Pan et al. [42]	2025	arXiv Preprint (25 pp.)	Data-centric LLM systems	×	×	✓	✓	29 prefill, 12 disagg; 0 carbon or energy

TABLE II
SYNTHESIS COMPARISON ACROSS EMPIRICAL SCHEDULING, LLM SERVING, AND LIFECYCLE CARBON CATEGORIES (L1.B–L1.F)

Category / Representative Works	Class	Scope	Phase	Grid CI	Emb.	KV WAN	Keep-Alive	Primary Knobs / Metrics
Wait Awhile [10], CarbonScaler [11], EcoVisor [30], Clover [39]	L1.B	Single DC	Non-LLM	✓	×	×	×	Temporal shift, MIG, batteries
GreenCourier [18], Sukprasert [12], Caribou [13], Calavaro [19]†	L1.B	Multi-DC	Non-LLM	✓	×	×	×	Geo-routing, HTTP/DAG hops, LP
Shahrad [31], CASA [22], EcoLife [23], LowCarb [32]†, LACE-RL [14]‡	L1.C	Serverl.	Non-LLM	✓	EcoLife	×	✓	Dynamic TTL, water+CI, DQN
ServerlessLLM [24], Festina [17]‡	L1.C	Serverl.	Agnostic/P/D	×	×	×	✓	NVMe Direct I/O, MPS, DVFS
vLLM [28], Sarathi [6], Wilkins [29], ThrottLL'eM [36]	L1.D	Engine	Agnostic/Col.	×	×	×	×	PagedAttn, chunking, NVML cap
DynamoLLM [7], TAPAS [37], Fernandez et al. [9]	L1.D	Cluster	Coloc. Phase	Dynamo	×	×	×	Multi-loop DVFS, TP, power cap
Splitwise [4], DistServe [5], Mooncake [15], DualScale [8]‡, Li [16]‡	L1.D	Intra-DC	Disagg. P/D	×	×	Intra-DC	×	P/D pools, 50% cap, MPC, RDMA
Gupta [2], [33]†, CarbonExplorer [34], LLM-Carbon [20], ISO [3], Sinha [35]†	L1.E	Lifecycle	Agnostic/Gen.	✓	✓	×	×	SCI Eq. (1), die/HBM capex
EcoServe [21]‡, Hewage et al. [38]	L1.E	Cluster	Agnostic	✓	✓	×	×	4R lifetimes (9/3 yr), NBTI aging
NIST [27], CloudSim [40]†, Azure Trace [26]	L1.F	Found.	Spec./Trace	×	×	×	×	Taxonomy, Cloud sim, conv/code logs
This Work (P4 Cluster + S3 Serverless)	–	**Multi-DC**	**Disagg. P/D**	✓	✓	✓	✓	**Hybrid A–D, SCI, GapHazard**

†Abstract-only verified (5 works in corpus). ‡Preprint / accepted conference preprint (EcoServe, DualScale, Li et al., LACE-RL, Festina).

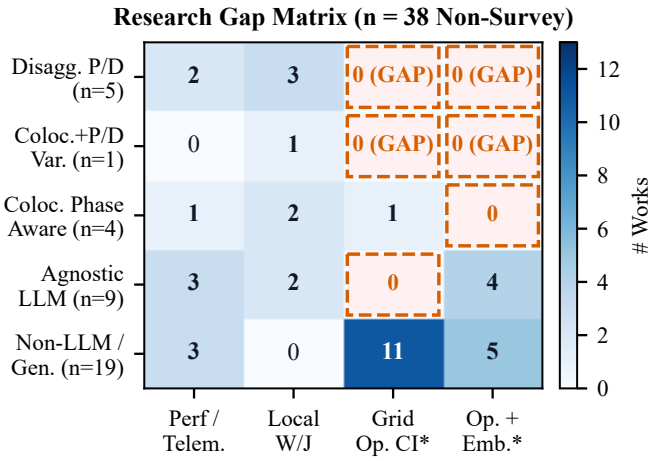


Fig. 3. Intersection of phase awareness and carbon/energy signal across $n = 38$ non-survey works. *Grid Op. CI includes 10 grid-CI systems plus 2 with battery/water co-signals [22], [30]; *Op. + Embodied includes 8 operational+embodied works plus CPU aging [38].

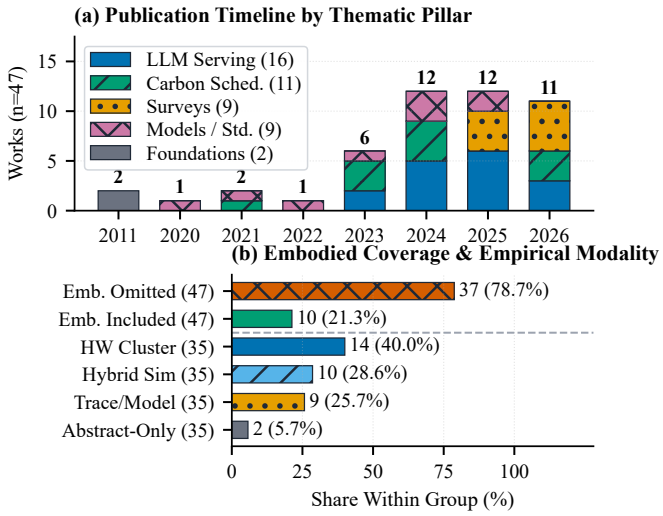


Fig. 4. Corpus bibliometrics from 2011 to 2026 ($n = 47$): (a) surge in 2024–2026 output (48.9% in 2025–2026); (b) 78.7% embodied omission and empirical evaluation split ($n = 35$).

across heterogeneous electricity grids that an independent researcher cannot instrument simultaneously. Model constants are separated into published hardware and serving baselines (Table III) and explicit calibration assumptions (Table IV), driven by public CC-BY-4.0 traces.

B. Phase Execution, Power, and Dispatch Formulation

For request i arriving at time t_i with prompt length $L_{\text{in}}(i)$ and output length $L_{\text{out}}(i)$, prefill duration on GPU tier g scales linearly above base kernel overhead $\tau_{P,0}(g)$:

$$T_{\text{exec},P}(i,g) = \tau_{P,0}(g) + \tau_{P,\text{tok}}(g) \cdot L_{\text{in}}(i), \tag{2}$$

calibrated to Splitwise Table IV uncontended Llama-70B anchors (84 ms conv and 95 ms code on H100; 155 ms and 185 ms on A100) [4]. During continuous-batching decode over active resident KV size K_{res} (GB) on G_r tensor-parallel GPUs, step time increases with HBM bandwidth contention:

$$t_{\text{step}}(g, K_{\text{res}}) = \tau_{D,0}(g)\left(1 + \alpha_{\text{mem}}(g)\frac{K_{\text{res}}}{G_r\,\text{BW}_{\text{HBM}}(g)}\right), \tag{3}$$

anchored to 28 ms on H100 and 42 ms on A100 [4], [7], with mild step inflation only if decode TDP cap $c_{\text{cap}} < 0.50$. Under colocation (`B1`), concurrent prefill slows active decode progress to factor $r_{\text{coloc}} = 0.20$ [4], [6], whereas disaggregation isolates decode iterations. End-to-end metrics aggregate cold-start delay, queueing, execution, KV transfer, and RTT:

$$\begin{aligned}\text{TTFT}_i &= T_{\text{cold},i} + T_{\text{q},P,i} + T_{\text{exec},P,i} + T_{\text{kv},i} + \text{RTT}_i,\\ \text{TPOT}_i &= \frac{T_{D,\text{exit},i} - T_{D,\text{enter},i}}{L_{\text{out}}(i)}.\end{aligned} \tag{4}$$

Instantaneous per-GPU power scales sublinearly with Model FLOPs Utilization (mfu) and active power cap ratio $c_{\text{cap}} \in (0,1]$ ($c_{\text{cap}} = 1.0$ on prefill; $c_{\text{cap}} = 0.55$ on decode [4]), plus GreenSKU Genoa host ancillary draw $P_{\text{host}}(\text{mfu})$:

$$\begin{aligned}P_{\text{gpu}}(\text{mfu}, c_{\text{cap}}) = P_{\text{idle}} + (c_{\text{cap}}P_{\text{TDP}} - P_{\text{idle}})\\ \times \left(\min\left(1.0, \frac{\text{mfu}}{\text{mfu}_{\text{sat}}}\right)\right)^{\gamma},\end{aligned} \tag{5}$$

using saturation threshold $\text{mfu}_{\text{sat}} = 0.45$ and calibrated sublinear exponent $\gamma = 0.70$.

For Llama-2-70B FP16 GQA-8 ($s_{\text{kv}} = 0.3125$ MB/tok), total KV payload $S_{\text{kv}}(i) = s_{\text{kv}}L_{\text{in}}(i)$ transfers over intra-DC NVLink/IB ($\delta_{\text{layer}} \in \{5, 8\}$ ms [4], $\text{BW}_{\text{local}} = 400$ Gbps) or cross-region WAN:

$$\begin{aligned}T_{\text{kv}}(r_P, r_D) &= \text{RTT}(r_P, r_D) + \frac{8S_{\text{kv}}}{B_{\text{wan}}},\\ E_{\text{kv}} &= P_{\text{net}}\frac{8S_{\text{kv}}}{3.6\times 10^6 B_{\text{wan}}}\,\text{PUE}.\end{aligned} \tag{6}$$

Cross-region pipelined disaggregation (Path C from r_P to $r_D \neq r_P$) is admissible only if both a latency recomputation crossover condition and an operational carbon crossover condition hold:

$$\begin{aligned}T_{\text{kv}}(r_P, r_D) + T_{\text{q},D}(r_D) < \min\big(T_{\text{q},D}(r_P),\\ T_{\text{recomp}}(L_{\text{in}}, r_D)\big),\end{aligned} \tag{7}$$

$$\begin{aligned}E_P I_{r_P}(t) + E_{\text{kv}}\frac{I_{r_P}(t) + I_{r_D}(t)}{2}\\ + E_D I_{r_D}(t) < \hat{C}_{\text{local}}(i).\end{aligned} \tag{8}$$

Across all feasible paths $p \in \{\text{A: Coloc., B: Intra P/D, C: Cross P/D, D: Remote Clean DC}\}$ meeting TTFT slack ($0.85 \cdot \text{TTFT}_{\text{SLO}}$), hybrid router `P4` selects:

$$\begin{aligned}p^* = \arg\min_{p\in\mathcal{F}(i,t)}\Big[w_c\hat{C}_{\text{op}}(p,t)\\ + 40\,w_q\max\big(0,\,\text{occ}_D(p) - 0.60\big)^2\Big],\end{aligned} \tag{9}$$

with $w_c = 1.0$, $w_q = 0.15$, and decode pool occupancy occ_D; when bursts exceed slack, `P4` falls back locally to Path A or Path B.

In Track B (serverless GPU serving), an online causal EWMA tracker (`OnlineGapHazardTracker`, `S3`) estimates discrete inter-arrival gap probabilities $\mathbf{p}$ across bins [0–1, 1–5, 5–10, 10–30, 30–60, $>$ 60] s and selects dynamic keep-alive timeout $k^* \in \{1, 5, 10, 30, 60\}$ s [14] at idle entry:

$$\begin{aligned}k^*(t) = \arg\min_{k\in\mathcal{K}}\Big[\mathbb{E}[\tau_{\text{idle}}(k)]\,r_{\text{idle}}(t)\\ + p_{\text{miss}}(k)\left(C_{\text{cold}}(t) + \frac{\beta_{\text{SLO}}\,\mathbb{I}_{\text{SLO}}}{\max(\widetilde{I}_r(t), 0.04)}\right)\Big],\end{aligned} \tag{10}$$

where $r_{\text{idle}}(t)$ is per-second warm-idle carbon rate ($\lambda_{\text{idle}} = 0.20$ baseline [14] or hardware idle ratio), $C_{\text{cold}}(t)$ is reload carbon during T_{cold}, and $\widetilde{I}_r(t)$ is normalized regional intensity.

Finally, operational emissions C_{op} integrate node and WAN power multiplied by PUE $= 1.12$ and regional grid CI $I_r(t)$, while amortized embodied emissions C_{emb} sum host and GPU footprints over lifetimes $\text{LT}_{\text{host}}, \text{LT}_{\text{gpu}}$ (4 yr uniform; sensitivity 9 yr host / 3 yr GPU [21]):

$$\begin{aligned}C_{\text{emb}} &= \sum_{m\in\mathcal{N}}\left(\frac{M_{\text{host},m}}{\text{LT}_{\text{host}}} + \frac{G_m M_{\text{gpu},m}}{\text{LT}_{\text{gpu}}}\right)T_{\text{sim}},\\ \text{SCI}_{\text{tot}} &= \frac{C_{\text{op}} + C_{\text{emb}}}{N_{\text{tok}}/1000}.\end{aligned} \tag{11}$$

C. Verified Baselines, Traces, and Calibrated Assumptions

Tables III and IV summarize verified profiles and calibrated assumptions. Multi-region cluster experiments (`E1`) span 3 zones (6 nodes, 24 H100s; 2 nodes/region): `South Scotland` (mean 9.75 gCO_2eq/kWh, 0–129), `London` (mean 163.32 gCO_2eq/kWh, 13–326), and `TEPCO` (mean 581.64 gCO_2eq/kWh, 302.3–671.5), from Oct. 2025 UK NESO 30-min traces (1,344 slots) and FY2024 Japan TSO traces (17,520 slots, converted from $kgCO_2/kWh$ via $\times$1,000). Workloads use Azure LLM conversational (`conv`: 19,366 reqs, 5.53 RPS, CV $= 1.094$, mean 1,154.7 in / 211.1 out) and coding (`code`: 8,819 reqs, 2.57 RPS, bursty CV $= 13.152$, mean 2,047.8 in / 27.9 out) traces [26] under DynamoLLM bucket SLOs (Short: 250/100; Medium: 400/100; Long: 2,000/100 ms TTFT/TPOT) [7].

Cluster runs compare five routing policies: `B1` (carbon-agnostic colocated round-robin), `B2` (intra-region P/D, 55% decode cap), `P3` (unconditional cross-region P/D when RTT $\leq$ TTFT slack), `P4` (hybrid Eqs. (7) through (9)), and `P5` (oracle 30-min grid lookahead).

TABLE III
VERIFIED HARDWARE, EMBODIED, AND LOADER PARAMETERS

Component	Verified Value(s)	Source
H100 SXM5 (4/node)	TDP 700 W, Idle 60 W, HBM 3352 GB/s, 124.9 $kgCO_2e$	[4], [21]
A100 SXM4 (4/node)	TDP 400 W, Idle 100 W, HBM 2039 GB/s, 121.5 $kgCO_2e$	[4], [21]
L4 Decode (8/node)	TDP 72 W, Idle 14.4 W, 300 GB/s, 41.2 $kgCO_2e$	[21]
GreenSKU Genoa	Peak 562.4 W, Idle 164.6 W, 921.6 $kgCO_2e$	[21], [26]
Decode Cap & KV	$c_{cap} = 0.55$; $\delta_{layer} = 5/8$ ms; PUE 1.12	[4]
70B / 13B Loaders	SLLM 10.3 / 2.0 s; Safe 48.0 / 8.2 s; PyT 84.0 / 14.0 s	[24]
Amortization Life	Uniform 4 yr; Asymmetric 9 yr host / 3 yr GPU	[2], [21]

TABLE IV
EXPLICIT SIMULATOR CALIBRATION ASSUMPTIONS (NON-MEASURED)

Parameter	Default / Range	Justification & Caveat
Synthetic WAN RTT	0.5 / 20 / 30 / 200 ms	Intra-DC / Intra-UK / Intra-JP / UK–JP delay
WAN KV BW	10 Gbps ({10, 25, 100})	Commodity Ethernet to DC WAN sweep
Sublinear MFU γ	0.70 ($mfu_{sat} = 0.45$)	Sublinear GPU load-power curve
Cold Power Frac.	0.65 P_{TDP} ([0.45, 0.85])	Deserialization + PCIe DMA warm-up
24h Extension	Empirical block tiling	Scales 58-min Azure blocks to 24h CI

Serverless runs (`E2`) compare `S1` (0 s), `S2` (30 s), `S2b` (60 s), and `S3` (GapHazard Eq. (10)) across `TEPCO`, `North Scotland` (mean 35.65 gCO_2eq/kWh, std 82.50, 0–355), and `South Scotland` using ServerlessLLM's 10.3 s 70B loader. All suites run across 10 seeds (100–109) and report Student-t 95% CIs ($t_{0.975,9} = 2.262$) and two-sided paired Wilcoxon p-values ($\alpha = 0.05$).

VI. SIMULATION RESULTS AND DISCUSSION

A. E1: Multi-Region Cluster Carbon–SLO Frontier

Table V and Fig. 5 summarize `E1` across 10 seeds (1h and 24h, conversational and code workloads). On conversational traffic (`conv`), hybrid routing `P4` reduces operational carbon intensity (SCI_{op}) by 60.8% (1h) and 60.7% (24h) relative to carbon-agnostic colocation `B1` (both $p = 0.002$), and by 43.9% (1h) and 44.1% (24h) relative to intra-region phase disaggregation `B2` ($p = 0.002$). Because operational savings far outweigh the fixed cluster embodied footprint ($SCI_{emb} \approx 0.0090$ $gCO_2eq/1k$ tok), total lifecycle intensity (SCI_{tot}) drops 54.6%–54.9% versus `B1` while maintaining 0.81%–0.82% SLO violations (TTFT P95 = 504.5–506.4 ms), well inside the 5% threshold.

By contrast, unconditional spatial disaggregation `P3` lowers SCI_{tot} slightly further (0.03858 at 24h) at the cost of breaching the 5% SLO boundary at 24h (5.90% violations, TTFT P95 = 1,190.4 ms); `P4` cuts `P3` violations by 83.6%–86.0% ($p = 0.002$) for a 6.5%–6.8% SCI_{tot} trade-off. Thus, **H1** and **H2** are supported on conversational serving.

Under extreme burstiness on coding completions (`code`, CV = 13.152, mean 2,047.8-token prompts), routing differences sharpen (Fig. 5b). `P4` cuts SCI_{op} by 28.5% versus `B1` and 23.4%–23.5% versus `B2` ($p = 0.002$). Meanwhile, `P3` triggers WAN KV stalls, driving SLO violations to 44.45%–44.57% (TTFT P95 = 30.7–39.3 s). `P4` prunes Path C (0%) via Eqs. (7) through (9), routes 14.11% onto Path A, 39.81% onto Path B, and 46.08% onto Path D (1h, Fig. 5c), and cuts SLO violations by 85.2%–86.0% (6.57% at 1h, 6.24% at 24h, $p = 0.002$) while matching `P3`'s SCI_{tot}.

Because `code` bursts also push `B1` (7.91%–8.46%) and `B2` (18.25%–19.47%) past 5%, **H1** is partially supported on `code` while **H2** is supported. Across all E1 suites, oracle lookahead `P5` tracks `P4` within 0.02% on carbon and 0.25% across all metrics ($p \geq 0.084$).

B. E2 & SW4: Serverless Keep-Alive and Loader Sensitivity

Table VI and Fig. 6 show that dynamic keep-alive efficacy depends on regional carbon intensity. On fossil-heavy `TEPCO` (581.6 gCO_2eq/kWh), 24h GapHazard `S3` achieves the lowest operational intensity ($SCI_{op} = 0.10825$), cutting carbon by 8.0% below scale-to-zero `S1` (0.11773), 7.0% below `S2` (30 s, 0.11644), and 11.0% below `S2b` (60 s, 0.12166, all $p = 0.002$). However, shortening timeouts during dirty slots raises SLO violations to 28.60% (4,529 cold starts) versus 19.45% under `S2` and 13.03% under `S2b` (still well below `S1`'s 40.25%); thus, **H3** is partially supported on `TEPCO`.

On clean `South Scotland` (9.75 gCO_2eq/kWh), inverse intensity scaling in Eq. (10) extends warm retention selectively, cutting SLO violations by 32.6% versus `S2` (12.99% vs. 19.29%, $p = 0.002$) while matching `S2b` (12.84%, $p = 0.125$) at a numerically lower operational cost (0.00094, $p = 0.084$ vs. S1/S2). Embodied emissions account for 97.4%–97.6% of total lifecycle SCI ($SCI_{emb} = 0.03308$), supporting **H4**. Volatile `North Scotland` shows a mixed split: `S3` cuts SLO violations by 30.5% versus `S2` ($p = 0.002$), yet zero-carbon wind slots yield paired zero carbon differences ($p \geq 0.50$).

In `SW4` (1h TEPCO, Fig. 6c), slow 70B loaders amplify `S3` savings: -23.7% vs. `S1` under PyTorch ($0.46686 \rightarrow 0.35601$ at 84 s) and -30.9% under Safetensors ($0.35452 \rightarrow 0.24488$ at 48 s), whereas `opt_13b` (2.0 s) yields lower carbon under `S1` (0.00831 vs. 0.00936, albeit 16.02% vs. 3.60% SLO violations). Applying H100's 8.6% hardware idle ratio (60 W / 700 W) instead of LACE-RL's 20% further drops `S3` SCI_{op} by 14.4% ($0.11193 \rightarrow 0.09578$).

C. Sensitivity Sweeps (`SW1`–`SW3`) and Clean Scale-Out Paradox (`H4`)

Fig. 7 shows operational limits across load, hardware tiering, and capacity:

1) **SW1 Load Scaling (0.5x–3.0x conv):** At 3.0x load, `P3` saturates clean pools (71.94% SLO viol., TTFT P95 = 446.7 s), `B1` stalls via colocation (54.48% viol., TPOT

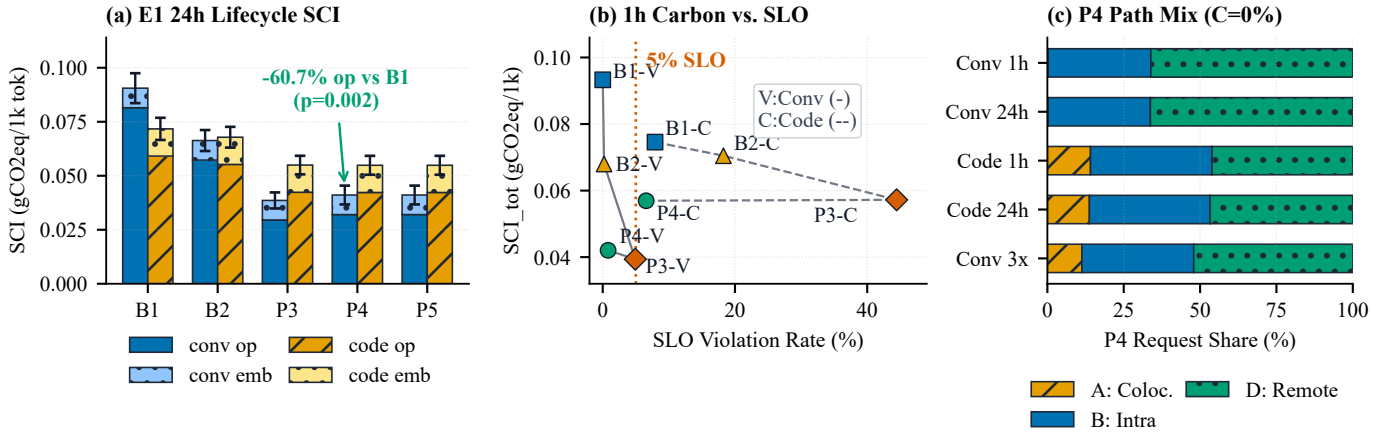


Fig. 5. Experiment `E1` cluster evaluation ($n = 10$ seeds, 95% Student-t CIs): (a) 24h operational plus embodied SCI; (b) 1h carbon-SLO frontier against the 5% violation boundary; (c) adaptive `P4` path distribution across regimes (Path C pruned to 0% at 10 Gbps WAN).

TABLE V
EXPERIMENT `E1` MULTI-REGION CLUSTER METRICS (MEAN ± 95% CI, $n = 10$ SEEDS; PAIRED WILCOXON p VS. `P4`)

Case	Policy	SCI_{op} (g/1k)	SCI_{tot} (g/1k)	TTFT P95 (ms)	SLO (%)	Wilcoxon p	Verdict (H1/H2)
1h Conv	`B1` (Coloc.)	0.08431 ± 0.00681	0.09329 ± 0.00681	268.6 ± 0.0	0.0005 ± 0.001	0.002 (op)	Baseline
1h Conv	`B2` (Intra P/D)	0.05901 ± 0.00472	0.06799 ± 0.00472	300.3 ± 0.2	0.20 ± 0.04	0.002 (op)	Baseline
1h Conv	`P3` (Uncond.)	0.03040 ± 0.00306	0.03938 ± 0.00306	1062.3 ± 136.1	4.93 ± 1.02	0.002 (SLO)	Tail Degrad.
1h Conv	`P4` (Hybrid)	**0.03307 ± 0.00365**	**0.04205 ± 0.00365**	**506.4 ± 2.1**	**0.81 ± 0.06**	–	**SUPPORTED**
24h Conv	`B1` / `B2`	0.08150 / 0.05726	0.09055 / 0.06631	268.6 / 300.3	0.00 / 0.20	0.002 (op)	Baselines
24h Conv	`P3` (Uncond.)	0.02953 ± 0.00369	0.03858 ± 0.00371	1190.4 ± 156.0	5.90 ± 1.10	0.002 (SLO)	Breaches 5%
24h Conv	`P4` (Hybrid)	**0.03203 ± 0.00434**	**0.04108 ± 0.00436**	**504.5 ± 3.7**	**0.82 ± 0.03**	–	**SUPPORTED**
1h Code	`B1` / `B2`	0.06182 / 0.05773	0.07457 / 0.07048	592.8 / 3209.1	7.91 / 18.25	0.002 (op)	CV 13.15
1h Code	`P3` (Uncond.)	0.04441 ± 0.00385	0.05722 ± 0.00385	39317.8 ± 793.9	44.45 ± 0.28	0.002 (SLO)	WAN Collapse
1h Code	`P4` (Hybrid)	**0.04419 ± 0.00388**	**0.05694 ± 0.00388**	**1613.8 ± 4.3**	**6.57 ± 0.25**	–	**H1 Part./H2 Sup.**
24h Code	`B1/B2/P3`	0.05911/0.05525/0.04235	0.07171/0.06784/0.05495	612.5/3380/30725	8.46/19.47/44.57	≤ 0.020	Baselines
24h Code	`P4` / `P5`	**0.04226** / 0.04226	**0.05485** / 0.05485	**1598.9** / 1599.0	**6.24** / 6.23	0.652–0.922	**H1 Part./H2 Sup.**

TABLE VI
EXPERIMENT `E2` 24H SERVERLESS KEEP-ALIVE METRICS ($n = 10$ SEEDS)

Region	Pol.	SCI_{op}	Emb. (%)	SLO (%)	Cold Starts
`TEPCO`	`S1` (0 s)	0.11773	22.2	40.25	6,271.6
`TEPCO`	`S2` (30 s)	0.11644	22.4	19.45	3,054.7
`TEPCO`	`S2b` (60 s)	0.12166	21.6	**13.03**	2,081.0
`TEPCO`	`S3` (GapHaz.)	**0.10825**	23.7	28.60	4,529.0
`N. Scot.`	`S1` (0 s)	0.00373	94.6	40.48	6,373.4
`N. Scot.`	`S2` (30 s)	0.00362	94.7	19.29	3,069.1
`N. Scot.`	`S2b` (60 s)	0.00378	94.6	**12.84**	2,073.4
`N. Scot.`	`S3` (GapHaz.)	**0.00323**	**95.0**	13.40	2,164.8
`S. Scot.`	`S1` (0 s)	0.00106	97.4	40.48	6,373.4
`S. Scot.`	`S2` (30 s)	0.00103	97.4	19.29	3,069.1
`S. Scot.`	`S2b` (60 s)	0.00108	97.4	**12.84**	2,073.4
`S. Scot.`	`S3` (GapHaz.)	**0.00094**	**97.6**	12.99	2,098.4

P95 = 2,331 ms), and `B2` reaches 20.00%. Activating 11.30% Path A spillover with Path B (36.62%) and Path D (52.08%), `P4` holds 2.89% violations ($SCI_{op} = 0.03483$ vs. `B1` 0.04943).

2) **SW2 Grid & Forecast Noise:** UK NESO forecasts, 30–60 min lags, and 2× noise shift `P4` SCI_{tot} by $< 0.3\%$. Under an intra-UK low-differential pair (`South Scotland` vs. `North West England`), `P4` holds 0.08% violations ($SCI_{op} = 0.00513$) vs. `P3`'s 10.71% (0.00425), a mixed arbitrage regime.

3) **SW3 Hardware Tiers & WAN:** Deploying `L4` decode nodes causes 24.73% (`B2`) and 53.47% (`P3`) decode collapse on `conv`, whereas `P4` holds 0.76% violations (Fig. 7b). Expanding WAN from 10 to 100 Gbps cuts `P3 code` violations from 44.45% to 20.78%, while `P4` stays at 6.57% because Eqs. (7) and (8) still reject cross-region KV forwarding (0% on `code`; 0.017% on 100 Gbps `conv`). At a strict 125 ms TTFT cap (`strict_code`), every policy exceeds 5% (`P4` 48.69%, `B1` 53.30%) due to WAN RTT and prompt latency floors.

4) **H4 Clean Scale-Out Paradox:** In 24h `E1`, clean `South Scotland` reaches a 93.0% embodied share on `code` (23.2% cluster-wide; 90.7% / 22.5% on `conv`). On all-UK (`South Scotland` + `North East` + `London`, where `P4_equal` has 53.3% embodied share), four spare clean nodes (`P3_overprov_plus4`, 10 vs. 6) trim SCI_{op} slightly (0.00804 vs. 0.00837) while raising SCI_{tot}

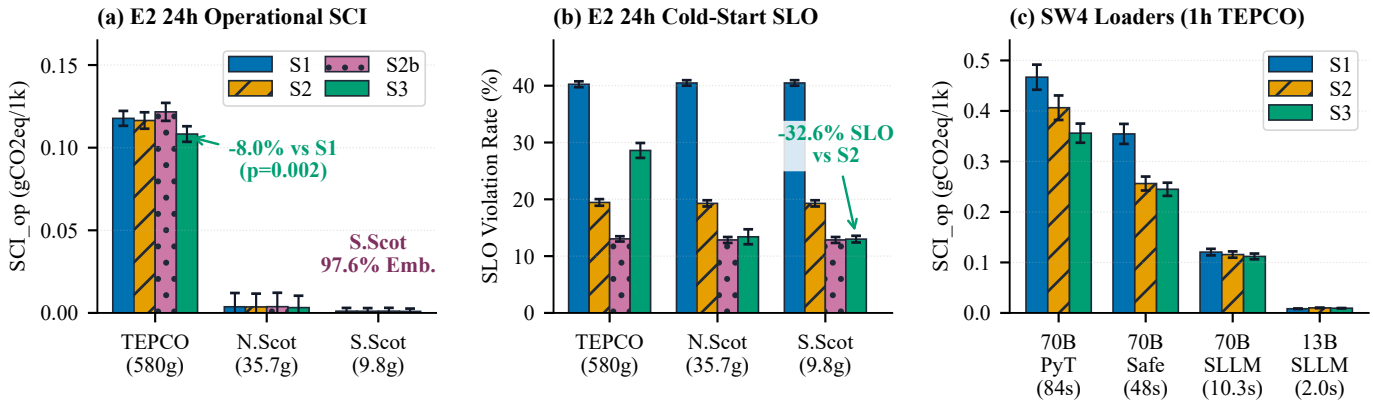


Fig. 6. Experiment `E2` and `SW4` serverless evaluation ($n = 10$ seeds, 95% CIs): (a) 24h operational SCI across grids; (b) 24h cold-start SLO violation rates showing regional carbon-SLO inversion; (c) `SW4` loader speedup impact (84 s PyTorch down to 10.3 s ServerlessLLM and 2.0 s 13B).

by +32.6% (0.02300 vs. 0.01735, 4-yr life) and +27.7% (0.01937 vs. 0.01516, 9-yr host / 3-yr GPU, Fig. 7c), supporting **H4**.

D. Answers to Research Questions (`RQ1–RQ4`)

- **RQ1 Answer:** Existing literature splits into grid-aware non-LLM cloud/serverless schedulers and carbon-agnostic LLM phase engines, with zero works (0/6) uniting P/D disaggregation with grid/embodied carbon and zero multi-DC carbon schedulers (0/4) modeling KV transfer economics, alongside 78.7% embodied omission.
- **RQ2 Answer:** Hybrid carbon-and-queue-aware phase routing (`P4`) lowers SCI_{op} by 60.7%–60.8% (SCI_{tot} by 54.6%–54.9%) over `B1` at $< 1\%$ SLO violations on conversational traffic, and cuts `P3` tail SLO violations by 85.2%–86.0% on bursty code completions through Path C pruning and Path A/B adaptive fallback.
- **RQ3 Answer:** GapHazard dynamic keep-alive (`S3`) cuts 24h operational carbon by 7.0%–11.0% on dirty grids (`TEPCO`) with a cold-start SLO trade-off (28.60% vs. 13.03% at 60 s TTL), shifts to SLO protection on clean grids (-32.6% SLO violations vs. 30 s TTL in `South Scotland`), and amplifies operational carbon gains to 23.7%–30.9% under slow 48–84 s loaders.
- **RQ4 Answer:** In wind-dominated regions (`South Scotland`), amortized hardware emissions account for 93.0% (cluster) to 97.6% (serverless) of total SCI. Overprovisioning clean GPU capacity raises total lifecycle SCI by 27.7%–32.6% across both uniform (4-yr) and decoupled (9-yr host / 3-yr GPU) amortization horizons.

VII. Open Challenges and Limitations

A. Synthesized Open Research Challenges (`OC1–OC7`)

Combining the literature gap matrix (Section IV) with the simulation trade-offs in Section VI identifies seven open research challenges:

- **OC1 (Cross-Region KV-Cache Transport):** Even at 100 Gbps WAN, the carbon-and-latency gate prunes cross-region P/D transfers for over 99.9% of arrivals. Closing that gap requires semantic token-level KV compression [46] integrated with live grid-carbon price gates.
- **OC2 (Live Marginal vs. Average Carbon Control):** Most cloud systems, including the baseline `E1` sweeps, optimize average regional grid intensity (gCO_2eq/kWh). Dispatching multi-megawatt LLM bursts requires marginal emission factors and grid-curtailment telemetry to reflect fossil peaker dispatch.
- **OC3 (Sub-Minute Embodied Cluster Sizing):** While per-request dispatch in `P4` minimizes operational carbon plus queue risk, `H4` shows that idle over-provisioning in clean zones reverses lifecycle gains (+32.6% SCI_{tot}). Joint control planes must couple sub-second routing with hourly embodied scale-down and heterogeneous reuse (9 yr/3 yr lifetimes [21], [38]).
- **OC4 (Serverless Multi-Tenant Checkpoint Sharing):** On `TEPCO`, dynamic keep-alive still faces an 8.0% vs. 28.60% carbon-SLO squeeze. Page-shared GPU weight caching across concurrent LoRA adapters and MPS partitions [17] can amortize 10.3 s NVMe cold starts across tenants.
- **OC5 (Water-Energy-Carbon Co-Optimization):** As CASA [22] notes for serverless CPUs, shifting LLM inference purely toward dry wind/solar regions risks shifting cooling water stress unless water intensity enters Eq. (1).
- **OC6 (Bursty Code-Completion Admission Control):** Under CV $= 13.152$, no fixed 6-node strategy stayed below 5% SLO violations without elastic spillover or prefill-priority admission pricing.
- **OC7 (Hardware Testbed Verification Across Continents):** Only 28.6% of reviewed empirical systems pair testbeds with trace simulators, and none tests live WAN disaggregated LLM serving across continents (0/6); multi-continent hardware validation remains untested.

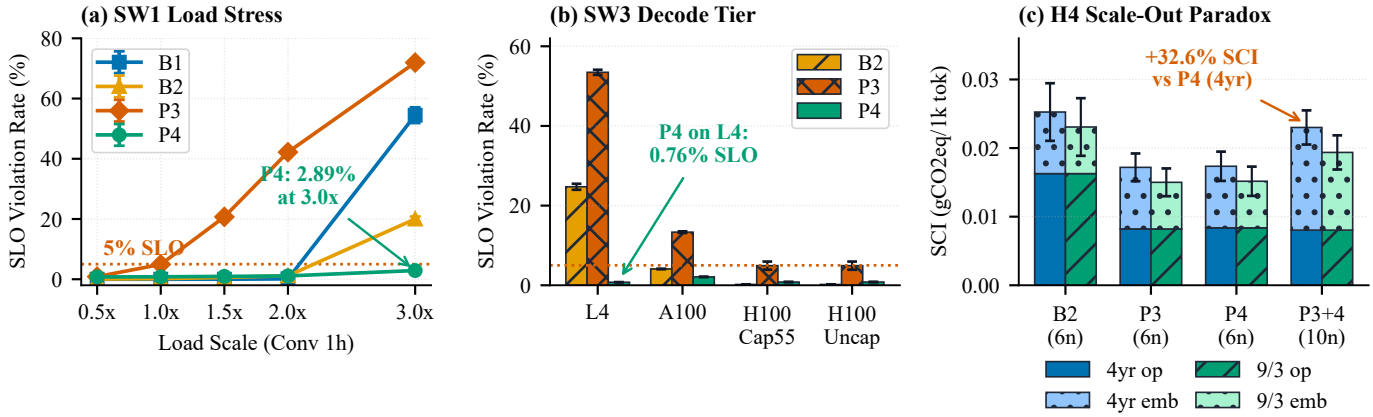


Fig. 7. Robustness and lifecycle evaluations ($n = 10$ seeds, 95% CIs): (a) `SW1` load scaling showing `P4` stability (2.89% at 3.0x) vs. P3/B1 collapse; (b) `SW3` decode GPU tiers (`P4` holds 0.76% on `L4`); (c) `H4` clean-region over-provisioning (+4 nodes) inflating SCI_{tot} by 27.7%–32.6%.

B. Limitations and Threats to Validity

Several methodological constraints bound the interpretation of these results:

1) **Simulation Abstraction vs. Physical Clusters:** Findings rely on analytical queueing and SimPy 4.1.2 discrete-event execution; no measurements come from instrumented multi-continent Kubernetes/vLLM deployments. Actual RDMA tail jitter, PCIe thermal throttling, and Linux memory fragmentation may shift absolute latency margins, although relative policy rankings (`P4` over `P3` and `B1`/`B2`) rest on structural bottlenecks observed in physical single-cluster studies [4], [5], [16].
2) **Calibrated Model Parameters:** Cross-region WAN RTT (20–200 ms), commodity WAN bandwidth (10–100 Gbps), sublinear MFU exponent ($\gamma = 0.70$), and cold-start GPU power fraction ($f_{\text{cold}} = 0.65$) are engineering assumptions bounded by published ranges (Table IV) and not direct hardware measurements; `SW3` and `SW4` sweep these parameters across wide intervals.
3) **24-Hour Trace Tiling:** Raw Azure LLM traces span 58.36 min (`conv`) and 57.27 min (`code`) [26]. Multi-day evaluations rely on block resampling across 24-hour UK/Japan grid curves while holding exact token distributions and arrival CVs constant.
4) **Live Dispatch vs. Provisioning Scope:** Inside Eq. (9), per-request `P4` routing scores operational cost and queue pressure; hardware embodied amortization is evaluated across fixed versus over-provisioned cluster configurations (`H4_elastic`) without dynamically powering servers on or off inside `E1`.
5) **Corpus Access and Preprints:** Five works (10.6%) could be verified only via abstracts [19], [32], [33], [35], [40], and eight full-text works (17.0%) are arXiv or accepted conference preprints (6 arXiv-only, 2 accepted preprints), which may undergo peer-reviewed revision prior to archival publication.

VIII. Conclusion

This paper bridged sustainable cloud scheduling and phase-disaggregated LLM serving through a PRISMA-2020 systematic synthesis of 47 works (42 full-text, 5 abstract-only) and a multi-seed SimPy 4.1.2 evaluation over real Azure, UK NESO, and Japan TSO traces. Literature cross-tabulation found zero existing coverage (0/6) uniting prefill/decode disaggregation with grid carbon or embodied accounting, zero multi-region carbon schedulers (0/4) modeling KV transfer economics, and 78.7% omission of manufacturing emissions.

Simulation results show that unconditional geographic carbon chasing (`P3`) causes severe tail-latency collapse on bursty code workloads (44.45% SLO violations) and load spikes (71.94% at 3.0x). By contrast, carbon-and-queue-aware hybrid routing (`P4`) preserves $< 1\%$ SLO violations on conversational traffic, lowers operational SCI by 60.7%–60.8% (-54.6%–54.9% lifecycle SCI vs. `B1`, $p = 0.002$), and cuts `P3` violations by 85.2%–86.0% on code traffic.

In serverless deployments, dynamic GapHazard keep-alive (`S3`) produces opposite trade-offs on dirty grids (-8.0% operational SCI on `TEPCO` with a 28.60% cold-start trade-off) and wind-clean grids (-32.6% SLO violations vs. 30 s static TTL in `South Scotland`), with operational carbon savings reaching 23.7%–30.9% under slower 48–84 s loaders. Finally, embodied hardware accounts for 93.0%–97.6% of lifecycle emissions in clean wind zones, causing four redundant nodes to raise total SCI by 27.7%–32.6%. Cloud operators should combine power-capped intra-region disaggregation with carbon-gated failover, fast NVMe checkpoint streaming, and utilization-matched clean capacity while avoiding idle over-provisioning in low-carbon regions.

Declaration of Generative AI Use

Generative AI tools were used during the preparation of this manuscript for literature search assistance, drafting, figure generation, and language editing. The author reviewed and verified all references, claims, data, and figures, and takes full responsibility for the content of this paper.

Data and Code Availability

All external workload and grid datasets are publicly available under CC-BY-4.0 licenses: Azure LLM Inference Traces and GreenSKU component tables via AzurePublicDataset (https://github.com/Azure/AzurePublicDataset and Zenodo `doi:10.5281/zenodo.10896254`), UK regional grid carbon intensity via the UK NESO Carbon Intensity API (https://api.carbonintensity.org.uk), and Japan FY2024 TSO emission factors via Zenodo (`doi:10.5281/zenodo.18396311`). The discrete-event simulator (SimPy 4.1.2), parameter manifests, PRISMA screening records, and figure-generation scripts are publicly available at https://github.com/UmarIqbal000/carbon-aware-llm-scheduling.

References

[1] J. Yang, Z. Saad, J. Wu, X. Niu, H. Leung, and S. Drew, "A survey on task scheduling in carbon-aware container orchestration," arXiv preprint arXiv:2508.05949v1, 2025, submitted to ACM Computing Surveys. [Online]. Available: https://arxiv.org/abs/2508.05949

[2] U. Gupta, Y. G. Kim, S. Lee, J. Tse, H.-H. S. Lee, G.-Y. Wei, D. Brooks, and C.-J. Wu, "Chasing carbon: The elusive environmental footprint of computing," in *2021 IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2021, pp. 854–867, arXiv:2011.02839. [Online]. Available: https://doi.org/10.1109/HPCA51647.2021.00076

[3] Green Software Foundation Standards Working Group, "Software carbon intensity (SCI) specification (version 1.1.0; ISO/IEC 21031:2024 basis)," Green Software Foundation / ISO/IEC 21031:2024 Standard Specification, 2024. [Online]. Available: https://sci.greensoftware.foundation/

[4] P. Patel, E. Choukse, C. Zhang, A. Shah, Í. Goiri, S. Maleki, and R. Bianchini, "Splitwise: Efficient generative LLM inference using phase splitting," in *2024 ACM/IEEE 51st Annual International Symposium on Computer Architecture (ISCA)*, 2024, pp. 118–132, arXiv:2311.18677. [Online]. Available: https://doi.org/10.1109/ISCA59077.2024.00019

[5] Y. Zhong, S. Liu, J. Chen, J. Hu, Y. Zhu, X. Liu, X. Jin, and H. Zhang, "DistServe: Disaggregating prefill and decoding for goodput-optimized large language model serving," in *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2024, arXiv:2401.09670. [Online]. Available: https://www.usenix.org/conference/osdi24/presentation/zhong-yinmin

[6] A. Agrawal, N. Kedia, A. Panwar, J. Mohan, N. Kwatra, B. Gulavani, A. Tumanov, and R. Ramjee, "Taming throughput-latency tradeoff in LLM inference with Sarathi-Serve," in *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2024, arXiv:2403.02310. [Online]. Available: https://www.usenix.org/conference/osdi24/presentation/agrawal

[7] J. Stojkovic, C. Zhang, Í. Goiri, J. Torrellas, and E. Choukse, "DynamoLLM: Designing LLM inference clusters for performance and energy efficiency," in *2025 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, 2025, pp. 1348–1362, arXiv:2408.00741. [Online]. Available: https://doi.org/10.1109/HPCA61900.2025.00102

[8] O. Basit, Y. Liu, Z. J. Kong, and Y. C. Hu, "DualScale: Energy-efficient disaggregated LLM serving via phase-aware placement and DVFS," arXiv preprint arXiv:2602.18755v3, 2026. [Online]. Available: https://arxiv.org/abs/2602.18755

[9] J. Fernandez, C. Na, V. Tiwari, Y. Bisk, S. Luccioni, and E. Strubell, "Energy considerations of large language model inference and efficiency optimizations," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL, Long Papers)*, vol. 1, 2025, pp. 32 556–32 569. [Online]. Available: https://doi.org/10.18653/v1/2025.acl-long.1563

[10] P. Wiesner, I. Behnke, D. Scheinert, K. Gontarska, and L. Thamsen, "Let's wait awhile: How temporal workload shifting can reduce carbon emissions in the cloud," in *Proceedings of the 22nd International Middleware Conference (Middleware)*, 2021, pp. 260–272, arXiv:2110.13234. [Online]. Available: https://doi.org/10.1145/3464298.3493399

[11] W. A. Hanafy, Q. Liang, N. Bashir, D. Irwin, and P. Shenoy, "CarbonScaler: Leveraging cloud workload elasticity for optimizing carbon-efficiency," *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, vol. 7, no. 3, pp. 1–28, 2023, arXiv:2302.08681. [Online]. Available: https://doi.org/10.1145/3626788

[12] T. Sukprasert, A. Souza, N. Bashir, D. Irwin, and P. Shenoy, "On the limitations of carbon-aware temporal and spatial workload shifting in the cloud," in *Proceedings of the Nineteenth European Conference on Computer Systems (EuroSys)*, 2024, pp. 924–941, arXiv:2306.06502. [Online]. Available: https://doi.org/10.1145/3627703.3650079

[13] V. U. Gsteiger, P. H. D. Long, Y. J. Sun, P. Javanrood, and M. Shahrad, "Caribou: Fine-grained geospatial shifting of serverless applications for sustainability," in *Proceedings of the ACM SIGOPS 30th Symposium on Operating Systems Principles (SOSP)*, 2024, pp. 403–420. [Online]. Available: https://doi.org/10.1145/3694715.3695954

[14] B. Sun, C. D. Antonopoulos, E. Smirni, B. Ren, N. Bellas, and S. Lalis, "Green or fast? learning to balance cold starts and idle carbon in serverless computing," in *Proceedings of the 26th IEEE International Symposium on Cluster, Cloud, and Internet Computing (CCGrid)*, 2026, accepted preprint, arXiv:2602.23935v1. [Online]. Available: https://arxiv.org/abs/2602.23935

[15] R. Qin, Z. Li, W. He, J. Cui, F. Ren, M. Zhang, Y. Wu, W. Zheng, and X. Xu, "Mooncake: Trading more storage for less computation — a KVCache-centric architecture for serving LLM chatbot," in *23rd USENIX Conference on File and Storage Technologies (FAST)*, 2025, arXiv:2407.00079. [Online]. Available: https://www.usenix.org/conference/fast25/presentation/qin

[16] J. Li, Y. Zhu, E. K. Lee, and K. Nahrstedt, "Revisiting disaggregated large language model serving for performance and energy implications," arXiv preprint arXiv:2601.08833v1, 2026. [Online]. Available: https://arxiv.org/abs/2601.08833

[17] T. Wang, G. Rattihalli, A. Dhakal, L. Shangguan, and D. Milojicic, "Energy-aware scheduling for serverless LLM serving on shared GPUs (extended)," in *Proceedings of the 2026 ACM SIGOPS Annual Technical Conference (SIGOPS ATC)*, 2026, accepted preprint, arXiv:2606.30391v2. [Online]. Available: https://arxiv.org/abs/2606.30391

[18] M. Chadha, T. Subramanian, E. Arima, M. Gerndt, M. Schulz, and O. Abboud, "GreenCourier: Carbon-aware scheduling for serverless functions," in *Proceedings of the 9th International Workshop on Serverless Computing (WoSC9)*, 2023, pp. 18–23, arXiv:2310.20375. [Online]. Available: https://doi.org/10.1145/3631295.3631396

[19] C. Calavaro, V. Cardellini, F. L. Presti, and G. R. Russo, "Carbon-aware offloading with function variants for serverless computing in the cloud-to-edge continuum," *Future Generation Computer Systems*, p. 108670, 2026. [Online]. Available: https://doi.org/10.1016/j.future.2026.108670

[20] A. Faiz, S. Kaneda, R. Wang, R. Osi, P. Sharma, F. Chen, and L. Jiang, "LLMCarbon: Modeling the end-to-end carbon footprint of large language models," in *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*, 2024, arXiv:2309.14393. [Online]. Available: https://arxiv.org/abs/2309.14393

[21] Y. Li, Z. Hu, E. Choukse, R. Fonseca, G. E. Suh, and U. Gupta, "EcoServe: Designing carbon-aware AI inference systems," arXiv preprint arXiv:2502.05043v2, 2025. [Online]. Available: https://arxiv.org/abs/2502.05043

[22] S. Qi, H. Moore, N. Hogade, D. Milojicic, C. Bash, and S. Pasricha, "CASA: A framework for SLO- and carbon-aware autoscaling and scheduling in serverless cloud computing," in *2024 IEEE 15th International Green and Sustainable Computing Conference (IGSC)*, 2024, pp. 1–6, arXiv:2409.00550. [Online]. Available: https://doi.org/10.1109/igsc64514.2024.00010

[23] Y. Jiang, R. B. Roy, B. Li, and D. Tiwari, "EcoLife: Carbon-aware serverless function scheduling for sustainable computing," in *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*, 2024, pp. 1–15, arXiv:2409.02085. [Online]. Available: https://doi.org/10.1109/sc41406.2024.00018

[24] Y. Fu, L. Xue, Y. Huang, A.-O. Brabete, D. Ustiugov, Y. Patel, and L. Mai, "ServerlessLLM: Low-latency serverless inference for large language models," in *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2024, arXiv:2401.14351. [Online]. Available: https://www.usenix.org/conference/osdi24/presentation/fu

[25] S. Alam, M. A. Ramzan, M. Zubair, U. Ullah, Ziaullah, R. Nawaz, and R. Ullah, "AI-driven resource allocation in cloud computing: a systematic review revealing critical sustainability and evaluation gaps,"

Computing, vol. 108, no. 5, p. 67, 2026, article 67. [Online]. Available: https://doi.org/10.1007/s00607-026-01655-8

[26] J. Stojkovic, C. Zhang, Í. Goiri, J. Torrellas, and E. Choukse, "Azure LLM inference trace 2024," Microsoft Azure Public Dataset (GitHub Repository, CC-BY; HPCA 2025 Companion Trace), 2024. [Online]. Available: https://github.com/Azure/AzurePublicDataset/blob/master/AzureLLMInferenceDataset2024.md

[27] P. Mell and T. Grance, "The NIST definition of cloud computing," National Institute of Standards and Technology (NIST), Special Publication 800-145, 2011. [Online]. Available: https://doi.org/10.6028/NIST.SP.800-145

[28] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. Gonzalez, H. Zhang, and I. Stoica, "Efficient memory management for large language model serving with PagedAttention," in *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023, pp. 611–626, arXiv:2309.06180. [Online]. Available: https://doi.org/10.1145/3600006.3613165

[29] G. Wilkins, S. Keshav, and R. Mortier, "Offline energy-optimal LLM serving: Workload-based energy models for LLM inference on heterogeneous systems," *ACM SIGEnergy Energy Informatics Review*, vol. 4, no. 5, pp. 113–119, 2024, arXiv:2407.04014. [Online]. Available: https://doi.org/10.1145/3727200.3727217

[30] A. Souza, N. Bashir, J. Murillo, W. Hanafy, Q. Liang, D. Irwin, and P. Shenoy, "Ecovisor: A virtual energy system for carbon-efficient applications," in *Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, vol. 2, 2023, pp. 252–265, arXiv:2210.04951. [Online]. Available: https://doi.org/10.1145/3575693.3575709

[31] M. Shahrad, R. Fonseca, Í. Goiri, G. Chaudhry, P. Batum, J. Cooke, E. Laureano, C. Tresness, M. Russinovich, and R. Bianchini, "Serverless in the wild: Characterizing and optimizing the serverless workload at a large cloud provider," in *2020 USENIX Annual Technical Conference (USENIX ATC)*, 2020, pp. 205–218. [Online]. Available: https://www.usenix.org/conference/atc20/presentation/shahrad

[32] R. B. Roy and D. Tiwari, "LowCarb: Carbon-aware scheduling of serverless functions," in *2026 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, 2026. [Online]. Available: https://doi.org/10.1109/hpca68181.2026.11408586

[33] U. Gupta, M. Elgamal, G. Hills, G.-Y. Wei, H.-H. S. Lee, D. Brooks, and C.-J. Wu, "ACT: Designing sustainable computer systems with an architectural carbon modeling tool," in *Proceedings of the 49th Annual International Symposium on Computer Architecture (ISCA)*, 2022, pp. 784–799. [Online]. Available: https://doi.org/10.1145/3470496.3527408

[34] B. Acun, B. Lee, F. Kazhamiaka, K. Maeng, U. Gupta, M. Chakkaravarthy, D. Brooks, and C.-J. Wu, "Carbon explorer: A holistic framework for designing carbon aware datacenters," in *Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, vol. 2, 2023, pp. 118–132, arXiv:2201.10036. [Online]. Available: https://doi.org/10.1145/3575693.3575754

[35] P. Sinha, D. Liakopoulos, R. Li, and N. J. Yadwadkar, "The utilization fallacy and the real drivers of carbon-efficient inference serving," *ACM SIGEnergy Energy Informatics Review*, vol. 5, no. 2, 2025. [Online]. Available: https://doi.org/10.1145/3757892.3757903

[36] A. K. Kakolyris, D. Masouros, P. Vavaroutsos, S. Xydis, and D. Soudris, "throttLL'eM: Predictive GPU throttling for energy efficient LLM inference serving," in *2025 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, 2025, arXiv:2408.05235. [Online]. Available: https://doi.org/10.1109/hpca61900.2025.00103

[37] J. Stojkovic, C. Zhang, Í. Goiri, E. Choukse, H. Qiu, R. Fonseca, J. Torrellas, and R. Bianchini, "TAPAS: Thermal- and power-aware scheduling for LLM inference in cloud platforms," in *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, vol. 2, 2025, arXiv:2501.02600. [Online]. Available: https://doi.org/10.1145/3676641.3716025

[38] T. B. Hewage, S. Ilager, M. R. Read, and R. Buyya, "Aging-aware CPU core management for embodied carbon amortization in cloud LLM inference," in *Proceedings of the 16th ACM International Conference on Future and Sustainable Energy Systems (ACM e-Energy)*, 2025, arXiv:2501.15829. [Online]. Available: https://doi.org/10.1145/3679240.3734608

[39] B. Li, S. Samsi, V. Gadepally, and D. Tiwari, "Clover: Toward sustainable AI with carbon-aware machine learning inference service," in *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC23)*, 2023, arXiv:2304.09781. [Online]. Available: https://doi.org/10.1145/3581784.3607034

[40] R. N. Calheiros, R. Ranjan, A. Beloglazov, C. A. F. D. Rose, and R. Buyya, "CloudSim: a toolkit for modeling and simulation of cloud computing environments and evaluation of resource provisioning algorithms," *Software: Practice and Experience*, vol. 41, no. 1, pp. 23–50, 2011. [Online]. Available: https://doi.org/10.1002/spe.995

[41] T. B. Hewage, S. Ilager, M. R. Read, P. Arroba, and R. Buyya, "Carbon-aware resource management for latency-sensitive cloud computing environments: A taxonomy and future directions," arXiv preprint arXiv:2609.03270v2, 2026. [Online]. Available: https://arxiv.org/abs/2609.03270

[42] J. Pan and G. Li, "A survey of LLM inference systems," arXiv preprint arXiv:2506.21901v1, 2025. [Online]. Available: https://arxiv.org/abs/2506.21901

[43] S. Agarwal, M. Rodriguez, and R. Buyya, "Dynamic function configuration and its management in serverless computing: A taxonomy and future directions," *ACM Computing Surveys*, vol. 58, no. 14, pp. 1–35, 2026. [Online]. Available: https://doi.org/10.1145/3821534

[44] V. Besozzi, M. D. Bartola, P. Dazzi, and M. Danelutto, "High-performance serverless computing: A systematic literature review on serverless for HPC, AI, and big data," *IEEE Access*, vol. 13, pp. 195 611–195 656, 2025. [Online]. Available: https://doi.org/10.1109/access.2025.3633989

[45] X. Miao, G. Oliaro, Z. Zhang, X. Cheng, H. Jin, T. Chen, and Z. Jia, "Towards efficient generative large language model serving: A survey from algorithms to systems," *ACM Computing Surveys*, vol. 58, no. 1, pp. 1–37, 2025. [Online]. Available: https://doi.org/10.1145/3754448

[46] J. Jiang, P. Yang, R. Zhang, and F. Liu, "Towards efficient large language model serving: A survey on system-aware KV cache optimization," in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, pp. 38 450–38 476. [Online]. Available: https://doi.org/10.18653/v1/2026.findings-acl.1916

[47] G. Bai, Z. Chai, C. Ling, S. Wang, J. Lu, N. Zhang, T. Shi, Z. Yu, M. Zhu, Y. Zhang, X. Song, C. Yang, Y. Cheng, and L. Zhao, "Beyond efficiency: A systematic survey of resource-efficient large language models," *ACM Computing Surveys*, vol. 58, no. 16, pp. 1–45, 2026. [Online]. Available: https://doi.org/10.1145/3845797